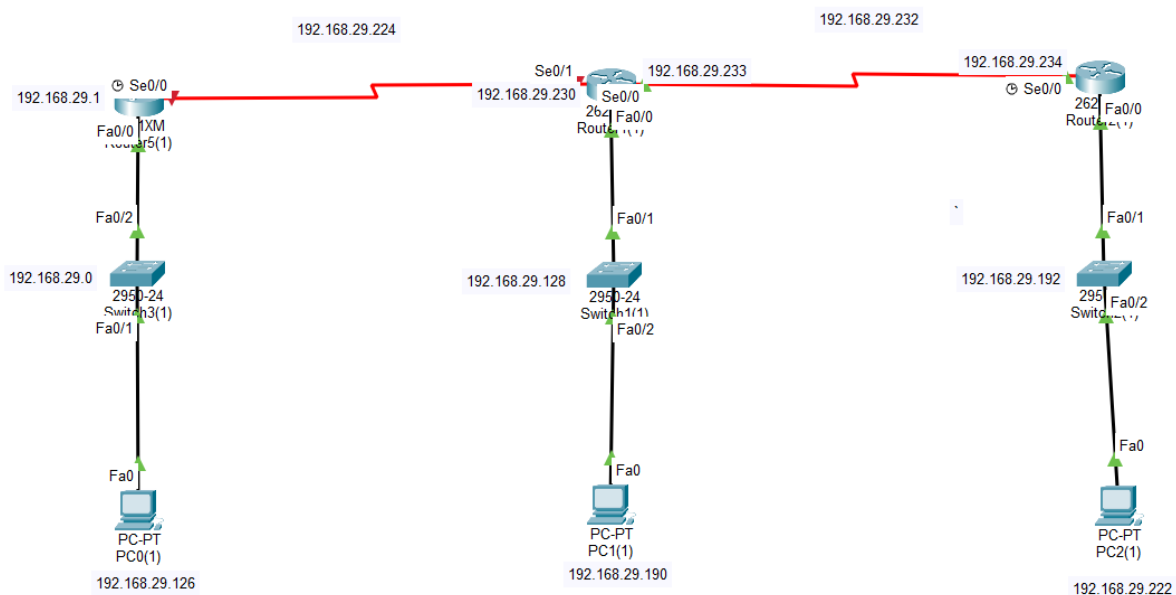


Day 3: Configuring RIP Theory: RIP concepts: Hop count, updates, and convergence
Practical : Configure RIP on a multi-router network. Task: Capture RIP updates using simulation tools and analyze them.

Task 1.

Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.



```
router rip
version 2
net 192.168.29.0    255.255.255.128
net 192.168.29.128 255.255.255.192
net 192.168.29.192 255.255.255.224
net 192.168.29.224 255.255.255.248
net 192.168.29.232 255.255.255.252

ex
do wr
```

Task 2.

Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.

Router 1:

```
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/25 is subnetted, 1 subnets
C       192.168.29.0 is directly connected, FastEthernet0/0

Router#
```

Router 2:

```
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/24 is variably subnetted, 3 subnets, 3 masks
C       192.168.29.128/26 is directly connected, FastEthernet0/0
R       192.168.29.192/27 [120/1] via 192.168.29.234, 00:00:26, Serial0/0
C       192.168.29.232/29 is directly connected, Serial0/0

Router#
```

Router 3:

```
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/24 is variably subnetted, 4 subnets, 4 masks
R       192.168.29.128/26 [120/1] via 192.168.29.233, 00:00:17, Serial0/0
C       192.168.29.192/27 is directly connected, FastEthernet0/0
R       192.168.29.224/29 [120/1] via 192.168.29.233, 00:00:17, Serial0/0
C       192.168.29.232/30 is directly connected, Serial0/0

Router#
```

Task 3.

Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables.

```
Router>en
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/24 is variably subnetted, 3 subnets, 3 masks
R       192.168.29.0/25 [120/1] via 192.168.29.225, 00:00:03, Serial0/1
C       192.168.29.128/26 is directly connected, FastEthernet0/0
C       192.168.29.224/29 is directly connected, Serial0/1

Router#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router#
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/24 is variably subnetted, 5 subnets, 4 masks
R       192.168.29.0/25 [120/1] via 192.168.29.225, 00:00:24, Serial0/1
C       192.168.29.128/26 is directly connected, FastEthernet0/0
R       192.168.29.192/27 [120/1] via 192.168.29.234, 00:00:01, Serial0/0
C       192.168.29.224/29 is directly connected, Serial0/1
C       192.168.29.232/29 is directly connected, Serial0/0
```

Task 4.

Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers.

Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.

The image displays a Cisco Packet Tracer simulation environment. On the left, a network diagram shows three routers (R1, R2, R3) connected in a line. R1 is connected to R2 via S0/0/0 (198.168.29.224) and S0/0/1 (198.168.29.232). R2 is connected to R3 via S0/0/0 (198.168.29.128) and S0/0/1 (198.168.29.192). Each router has a Fa0/24 interface connected to a switch (S1, S2, S3) and a Fa0/24 interface connected to a PC (PC1, PC2, PC3). The switches are connected to the routers via Fa0/24 and Fa0/24. The PCs are connected to the switches via Fa0/24 and Fa0/24.

On the right, the Event List panel shows a list of events. The first event is a STP packet from Switch2(1) to Router2(1) at 0.692 seconds. The second event is a STP packet from Router2(1) to PC2(1) at 0.693 seconds. The third event is a STP packet from PC2(1) to Switch3(1) at 0.693 seconds. The fourth event is a STP packet from Switch3(1) to Router1(1) at 0.696 seconds. The fifth event is a STP packet from Router1(1) to Router2(1) at 0.697 seconds. The sixth event is a STP packet from Router2(1) to Switch2(1) at 0.698 seconds. The seventh event is a STP packet from Switch2(1) to PC2(1) at 0.699 seconds. The eighth event is a STP packet from PC2(1) to Switch2(1) at 0.700 seconds. The ninth event is a STP packet from Switch2(1) to Router2(1) at 0.701 seconds. The tenth event is a STP packet from Router2(1) to Router1(1) at 0.702 seconds. The eleventh event is a STP packet from Router1(1) to Router5(1) at 0.703 seconds. The twelfth event is a STP packet from Router5(1) to Switch3(1) at 0.704 seconds. The thirteenth event is a STP packet from Switch3(1) to PC0(1) at 0.705 seconds. The fourteenth event is a STP packet from PC0(1) to Switch1(1) at 0.756 seconds. The fifteenth event is a STP packet from Switch1(1) to Router1(1) at 0.757 seconds. The sixteenth event is a STP packet from Router1(1) to PC1(1) at 0.757 seconds. The seventeenth event is a STP packet from PC1(1) to Switch3(1) at 0.758 seconds. The eighteenth event is a STP packet from Switch3(1) to Router5(1) at 0.759 seconds. The nineteenth event is a STP packet from Router5(1) to PC0(1) at 0.759 seconds. The twentieth event is a STP packet from PC0(1) to Switch3(1) at 1.709 seconds. The twenty-first event is a STP packet from Switch3(1) to Router5(1) at 1.710 seconds. The twenty-second event is a STP packet from Router5(1) to Router1(1) at 1.711 seconds. The twenty-third event is a STP packet from Router1(1) to Router2(1) at 1.712 seconds. The twenty-fourth event is a STP packet from Router2(1) to Switch2(1) at 1.713 seconds. The twenty-fifth event is a STP packet from Switch2(1) to PC2(1) at 1.714 seconds. The twenty-sixth event is a STP packet from PC2(1) to Switch2(1) at 1.715 seconds. The twenty-seventh event is a STP packet from Switch2(1) to PC2(1) at 1.716 seconds.

The Simulation Panel shows the Event List with columns: Vis., Time(sec), Last Device, At Device, and Type. The events are filtered for STP packets. The first event is at 0.702 seconds, from Router2(1) to Router1(1), type ICMP. The second event is at 0.703 seconds, from Router1(1) to Router5(1), type ICMP. The third event is at 0.704 seconds, from Router5(1) to Switch3(1), type ICMP. The fourth event is at 0.705 seconds, from Switch3(1) to PC0(1), type ICMP. The fifth event is at 0.756 seconds, from -- to Switch1(1), type STP. The sixth event is at 0.757 seconds, from Switch1(1) to Router1(1), type STP. The seventh event is at 0.757 seconds, from Switch1(1) to PC1(1), type STP. The eighth event is at 0.758 seconds, from -- to Switch3(1), type STP. The ninth event is at 0.759 seconds, from Switch3(1) to Router5(1), type STP. The tenth event is at 0.759 seconds, from Switch3(1) to PC0(1), type STP. The eleventh event is at 1.709 seconds, from -- to PC0(1), type ICMP. The twelfth event is at 1.710 seconds, from PC0(1) to Switch3(1), type ICMP. The thirteenth event is at 1.711 seconds, from Switch3(1) to Router5(1), type ICMP. The fourteenth event is at 1.712 seconds, from Router5(1) to Router1(1), type ICMP. The fifteenth event is at 1.713 seconds, from Router1(1) to Router2(1), type ICMP. The sixteenth event is at 1.714 seconds, from Router2(1) to Switch2(1), type ICMP. The seventeenth event is at 1.715 seconds, from Switch2(1) to PC2(1), type ICMP. The eighteenth event is at 1.716 seconds, from PC2(1) to Switch2(1), type ICMP.

The Simulation Panel also includes a Reset Simulation button, a Constant Delay checkbox, and a Play Controls section with buttons for Play, Pause, and Stop. The Event List Filters - Visible Events section lists various protocols and services, including ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP.